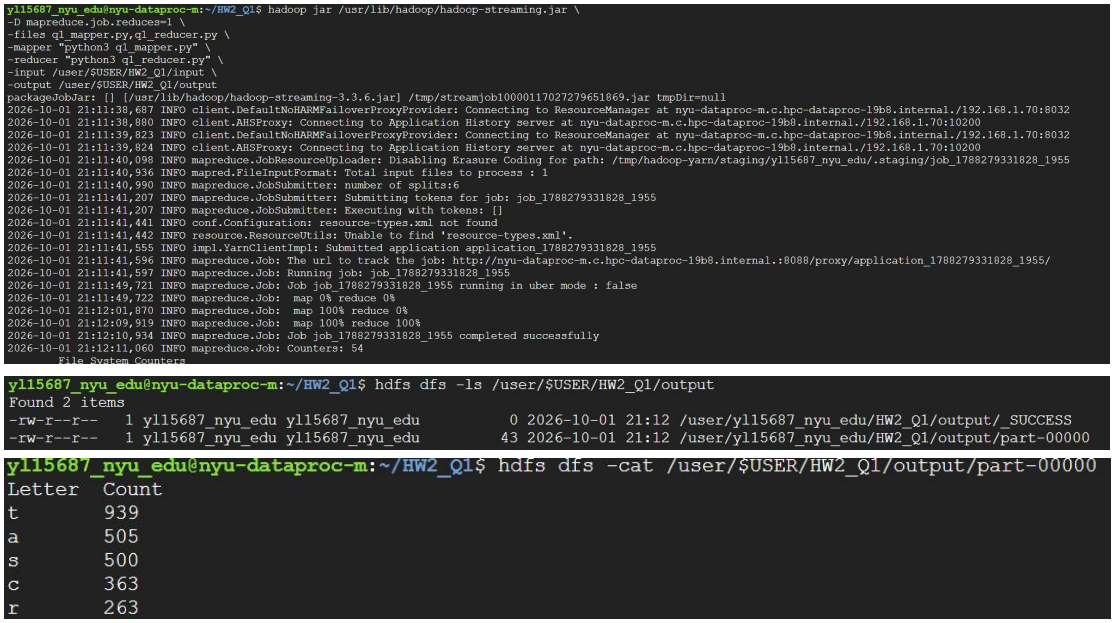
## Screenshot:










## Text:

hadoop jar /usr/lib/hadoop/hadoop-streaming.jar \
-D mapreduce.job.reduces=1 \
-files q1_mapper.py,q1_reducer.py \
-mapper "python3 q1_mapper.py" \
-reducer "python3 q1_reducer.py" \
-input /user/$USER/HW2_Q1/input \
-output /user/$USER/HW2_Q1/output

## Reflection:

1. What did you learn from this assignment, and what was the most challenging aspect of it?

I learned how a mapper and reducer work together in Hadoop Streaming. The mapper processed each word and output its first letter, while the reducer counted the letters and found the Top 5.

The most challenging part was cleaning the text correctly, especially handling punctuation, numbers, hyphenated words, and apostrophes.

2. How did you approach and resolve that challenge?

I used a regular expression in the mapper to extract valid words. I also converted the text to lowercase so uppercase and lowercase words were counted together. By the way, I tested the mapper with examples " before running the Hadoop job.